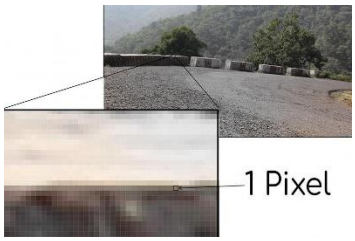
Gate 1 — How does a computer see?

1. What is a digital image? Explain a pixel, image resolution, and the matrix representation of an image. 2. What are RGB and grayscale images? Why might an autonomous vehicle process them differently?

1.1 What is a digital image?

A computer does not see a road as we do. It stores an image as a collection of tiny elements called **pixels**.



Think of a digital image as a table of numbers.

For a grayscale image, a pixel typically has an intensity value between 0 and 255:

- 0 = black
- 255 = white
- Values in between = shades of grey

For example, a tiny grayscale image might be represented as:

$$I = \begin{bmatrix} 0 & 50 & 100 \\ 120 & 180 & 220 \\ 255 & 200 & 80 \end{bmatrix}$$

Each number represents one pixel. The matrix has 3 rows and 3 columns.

Image resolution tells you how many pixels an image contains. An image with width 640 and height 480 contains $640 \times 480 = 307{,}200$ pixels.

A higher resolution can preserve more visual detail, but it also means more data to process.

1.2 RGB versus grayscale

RGB image: Each pixel stores red, green and blue intensity values. Useful for color-based features

Grayscale image : Each pixel stores one brightness value. Often useful for edge detection and simpler processing.





Why does this matter for lane detection?

A white lane marking is usually brighter than the road surface. Grayscale processing makes that brightness difference easier to analyze without having to process three color channels.

However, grayscale loses color information, which can sometimes be useful when the road and its surroundings have similar brightness. An autonomous vehicle processes RGB and grayscale images differently because of differences in data volume, computational load, and semantic information.

Gate 2 — Basic image processing

1. What is Gaussian blur and why is it useful before detecting features? 2. What is HSV colour space? Why can it sometimes be more useful than RGB for road/lane detection? 3. What does Canny edge detection do? 4. What is a Region of Interest (ROI), and why can it reduce false detections?

2.1 Gaussian blur

Real road images contain noise: small textures, gravel, shadows and tiny intensity variations. If you detect edges immediately, you may detect too many irrelevant details.

Gaussian blur smooths an image by replacing each pixel with a weighted average of nearby pixels. Nearby pixels receive weights based on a Gaussian distribution.





A kernel is a small matrix of numbers that you move over an image to process it. At each position, the kernel interacts with the corresponding image pixels to calculate a new pixel value. smoothing kernel is a kernel designed to reduce small variations and noise in an image. Gaussian kernel is a special type of smoothing kernel.

The difference is how it assigns weights to the pixels around the center.

- An ordinary averaging kernel gives equal weight to all pixels in its window.
- A Gaussian kernel gives greater weight to pixels near the center and smaller weights to pixels farther away.

For example, a Gaussian kernel of size : $I = 1/16 \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

2.2 HSV colour space

RGB represents colors using red, green and blue. HSV represents them using:

- **H — Hue:** the type of color.
- **S — Saturation:** how vivid or grayish the color is.

- **V — Value:** brightness.

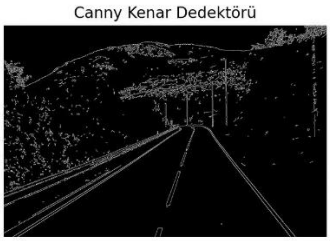
HSV can be useful when you want to isolate yellow lane markings or bright white road markings. You can select ranges of color or brightness without setting independent thresholds for all three RGB channels.

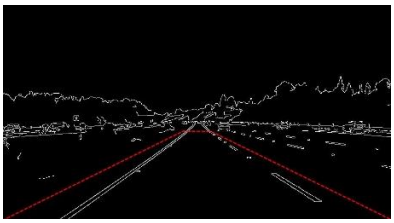
2.3 Canny edge detection

Imagine tracing the boundaries of objects in a photograph. Canny attempts to find locations where image intensity changes sharply.

For lane detection, these edges may correspond to the borders of painted lane markings.







The output is usually a binary image:

- Black pixels = no detected edge.
- White pixels = detected edge.

Canny does **not** know which edges are lane lines. It may also detect the edges of cars, trees, road cracks and shadows. We need more steps to filter those out.

2.4 Region of Interest (ROI)

In a forward-facing road image, the sky and most buildings are irrelevant to the lane directly in front of the car. A **Region of Interest** restricts processing to a selected part of the image, often a polygon around the road.

For example, imagine selecting a polygon whose vertices are near the bottom-left, upper-middle, bottom-right and bottom of the image. This roughly covers the road ahead.

Key idea: ROI reduces irrelevant edges, but cannot guarantee that every remaining edge belongs to a lane. After applying Canny, the computer may detect edges from:

- Lane markings ✅
- Trees and buildings ❌
- Vehicles ❌
- Road cracks and texture ❌

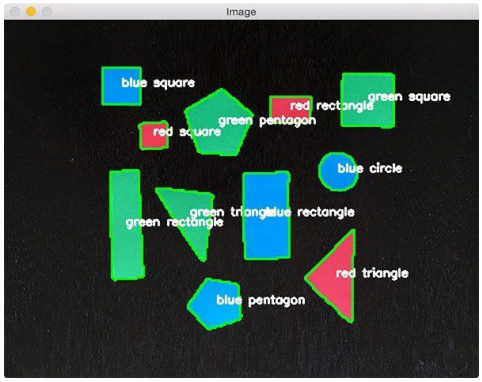
Canny detects edges based on intensity changes. It doesn't know which edges belong to lanes. Hence roi helps to detect our path better.

Gate 3 — Detecting structure

1. What is a contour? What information can a contour provide? 2. What is the Hough Transform? Explain how it can be used to find lane lines

3.1 What is a contour?

A contour is a curve joining points along the boundary of a shape or connected region in an image.



Contours can help you estimate:

- The boundary and shape of an object.
- Its area and perimeter.
- Its bounding box and approximate position.
- Whether a connected region has a shape worth investigating.

.

3.2 What is the Hough Transform?

The **Hough Transform detects geometric shapes by collecting evidence from image points**. It can use its line-detection version.



Hough Transform works roughly like this:

1. Take an edge pixel.
2. Consider the different straight lines that could pass through it.
3. Each possible line gets a vote in a parameter space.
4. Repeat for other edge pixels.
5. Lines receiving enough votes become strong candidates.

Gate 4 — Lane / path perception

1. Why is lane/path detection useful for an autonomous ground vehicle? 2. Why can shadows, road texture, poor lighting, missing markings, or clutter create false detections? 3. Why is lane detection alone not sufficient for a fully autonomous vehicle?

4.1 Why is lane detection useful?

A lane detector estimates the road's lane boundaries from camera images. The vehicle can use this information to estimate its position relative to the lane and support steering and navigation.

It is useful because it helps the vehicle:

- Stay in its lane and maintain a suitable position on the road.
- Estimate the road's direction and curvature to plan steering.
- Avoid drifting outside the drivable area.
- Support path planning and control, helping the vehicle follow a safe route.

However, **lane detection alone is not enough to make a vehicle autonomous**. A complete system also needs obstacle detection, localization, planning, control, and handling of dynamic situations.

4.2 Why do lane detectors make mistakes?

Challenge	**Why it causes problems**
Shadows	They create sharp intensity changes that look like edges.
Road texture	Cracks and patches produce irrelevant lines.
Poor lighting	Lane markings may become difficult to distinguish.
Missing markings	One or both lane boundaries may not be visible.
Clutter	Cars, road signs and other objects create competing edges.

Lane detection provides information about the road, but it does not provide a complete understanding of the vehicle's surroundings.

A fully autonomous vehicle also needs to:

- Detect obstacles: Identify pedestrians, vehicles, cyclists, and debris.
- Understand traffic rules: Recognize traffic lights, signs, intersections, and right-of-way.
- Localize itself: Estimate its position and orientation in the environment.
- Plan safe movements: Decide when to turn, stop, change lanes, or avoid an obstacle.
- Control the vehicle: Convert planned movements into steering, acceleration, and braking commands.

Gate 5 — Perspective and Bird's-Eye View (BEV)

5.1 Perspective distortion

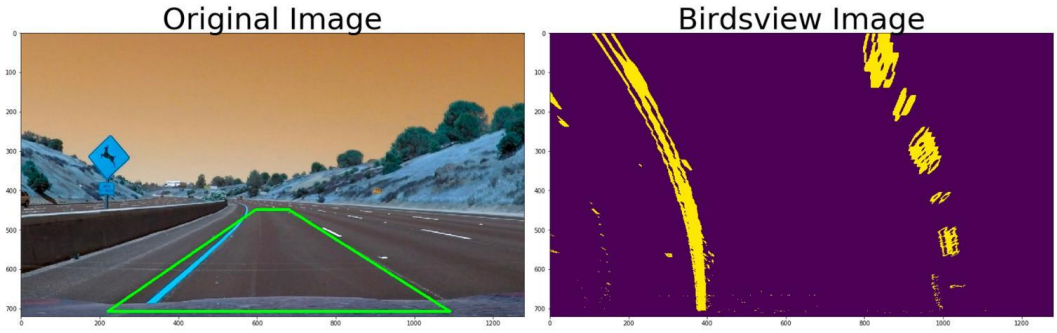
A camera sees the three-dimensional world projected onto a two-dimensional image. Objects that are farther away appear smaller, and parallel road boundaries can appear to converge toward a vanishing point.

This makes distances and road geometry difficult to judge directly from image pixels.

5.2 What is a Bird's-Eye View?







A Bird's-Eye View transforms a selected road region into an approximate top-down representation using a perspective transformation.

Why is this useful?

- Lane geometry can become easier to compare.
- The road's width and curvature can be easier to analyze.
- Navigation-related measurements may become more intuitive.